

ASHLEE FAROTTE-KRUCHAS

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EDUCATION

University of Oklahoma

Norman, OK

Dodge Family College of Arts and Sciences

- Bachelor of Science in Physics - in progress
- Bachelor of Science in Biology
- Minor in Mathematics
- Undergraduate Certificate in Data Analytics
- Overall GPA: 3.51

RESEARCH EXPERIENCE

Undergraduate Researcher - Computational Astrophysics

Norman, OK

University of Oklahoma

2026 - Present

Research Mentors: Dr. Nagaraj Vernekar and Dr. Michael Hayden

- Develop and train neural networks for stellar spectral modeling using Python, TensorFlow, and Keras.
- Work with large astronomical datasets and observed stellar spectra from surveys including DESI and APOGEE.
- Crossmatch stellar catalogs and prepare independent validation data for evaluating model performance.
- Use high-performance computing resources and SLURM workflows to run and monitor computational research jobs.
- Contribute to a project aimed at efficiently modeling lower-resolution spectra and enabling analysis of the large stellar datasets produced by modern astronomical surveys.

MUSEUM & BIOLOGICAL EXPERIENCE

Vertebrate Paleontology Preparation Lab Volunteer

Norman, OK

Sam Noble Museum

11/2023 - 12/2024

- Prepared macro- and microvertebrate fossils for scientific research through careful cleaning, sorting, stabilization, and specimen preparation.
- Created molds for incomplete or damaged specimens and followed detailed preparation instructions from laboratory supervisors.
- Used specialized hand and pneumatic tools and laboratory materials, including dental picks, air scribes, cast-cutter saws, acetone, and bonding materials.

School Aquarium Program Volunteer

Lompoc, CA

Cabrillo High School

08/2019 - 04/2020

- Supported a school aquarium program and provided educational tours for elementary students, senior citizens, students with special needs, and members of the public.
- Helped communicate biological and marine-science concepts to visitors through tours and educational activities.
- Performed aquarium husbandry and tank management, including cleaning and maintaining tanks, feeding animals, and monitoring animals for potential health concerns.

TECHNICAL & SCIENTIFIC SKILLS

Programming & Data: Python, MATLAB, NumPy, Pandas, HDF5, Astropy, scientific data processing, data visualization

Machine Learning & Computing: TensorFlow, Keras, neural networks, model training, high-performance computing (HPC), SLURM

Astronomical Research: Stellar spectroscopy, DESI, APOGEE, Gaia-based catalog crossmatching, spectral data validation

Laboratory & Museum: Fossil/specimen preparation, mold preparation, specimen handling, specialized hand and pneumatic tools

Animal Care & Outreach: Aquarium husbandry, tank maintenance, animal feeding and monitoring, educational tours, science outreach

SELECTED COURSEWORK

Computational Physics; Advanced Laboratory; Physical Mechanics; Electricity & Magnetism; Numerical Linear Algebra; Partial Differential Equations; Neurobiology; Neuropharmacology; Ichthyology; Animal Behavior; Genetics; Physiology; Evolution

ADDITIONAL WORK EXPERIENCE

Server	Midwest City, OK
<i>Olive Garden</i>	02/2023 - 03/2024
• Provided customer service in a fast-paced environment while coordinating orders, food service, and team responsibilities.	
Shift Lead Barista / Trainer	Midwest City, OK
<i>Scooter's Coffee</i>	05/2022 - 02/2023
• Completed shift-lead responsibilities, trained new employees, handled operational tasks, and supported efficient customer service.	